

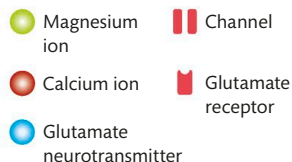
Networks in the Brain

The patterns of nerve-cell connections in the human brain are believed to influence how it processes sensory perceptions, performs cognitive tasks, and stores memories.

Wiring the brain

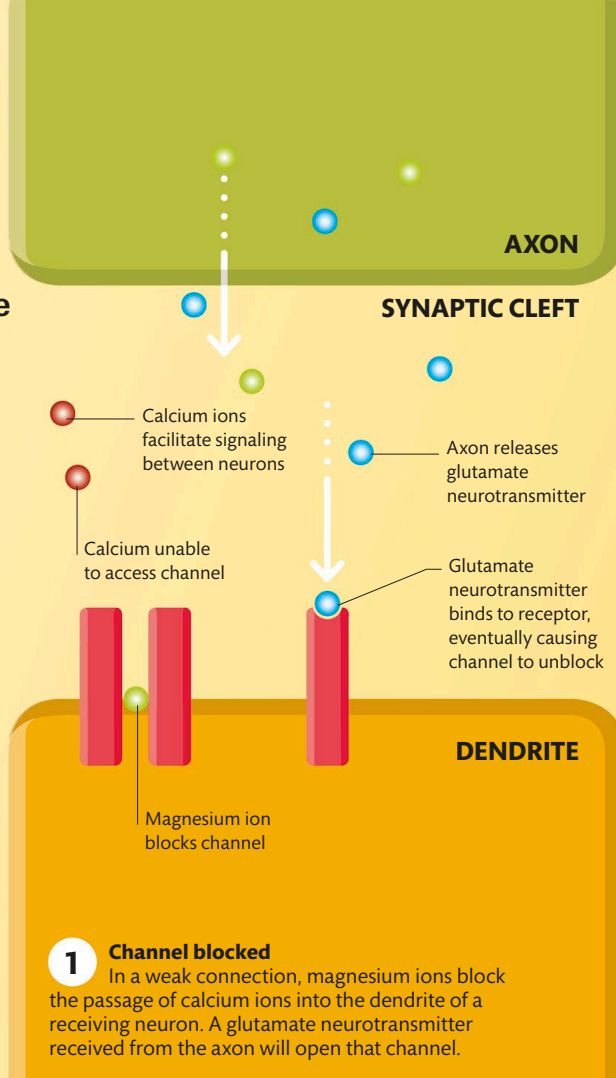
The dominant theory of how the brain remembers and learns can be summed up by the phrase “the cells that fire together, wire together.” It suggests that repeated communication between cells creates stronger connections between them, and a network of cells emerges in the brain that is associated with a specific mental process—such as a movement, a thought, or even a memory (see pp.136–37).

KEY



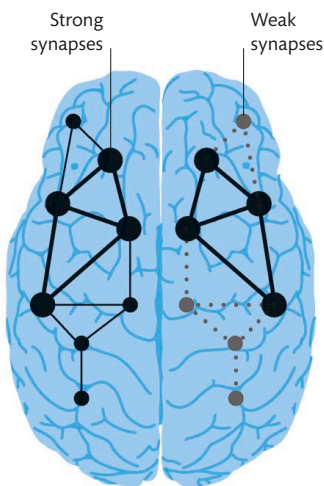
Synaptic weight

Little-used connections have channels blocked by magnesium ions. As the strength of a connection between two neurons in a network increases, the channel is unblocked, and the number of receptors at the synapse increases.



Neuroplasticity

The networks of the brain are not fixed but seem to change and adapt in accordance with mental and physical processes. This means that old circuits associated with one memory or a skill that is no longer in use fade in strength as the brain devotes attention to another and forms a new network with other cells. Neuroscientists say the brain is plastic, meaning its cells and the connections between them can be reformed many times over as required. Neuroplasticity allows brains to recover abilities lost due to brain damage.



BRAIN PATHWAYS

WHAT IS THE BRAIN'S DEFAULT MODE NETWORK?

It is a group of brain regions that show low activity levels when engaged in a task such as paying attention but high activity levels when awake and not engaged in a specific mental task.